BE700 Homework 3: Insulin Resistance, Diabetes and Aging in Human Skeletal Muscle

Analysis of GSE25462 (Jin et al., JCI 2011)

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1 Introduction and hypotheses

Insulin resistance in skeletal muscle precedes type 2 diabetes (T2D), and aging is itself a major risk factor for both. Jin et al. (2011) profiled vastus lateralis biopsies from normoglycemic people with or without a family history (FH) of T2D and from people with T2D, and reported that increased serum response factor (SRF) activity marks insulin resistance. The Hallmarks of Aging (López-Otín et

al., 2013) predicts that age-associated changes should involve mitochondrial dysfunction, deregulated nutrient sensing, altered intercellular communication (inflammation), and loss of proteostasis.

Hypotheses tested here:

- **H1.** Unsupervised clustering of muscle transcriptomes separates samples by a biological factor (sex, diabetes status, or age), not at random.
- **H2.** Sex is the strongest single source of transcriptome-wide variation, driven by sex-chromosome genes.
- **H3.** A set of genes is correlated with age, and these genes are enriched for Hallmark-of-Aging processes (mitochondria / oxidative phosphorylation, extracellular matrix, inflammation).
- **H4.** Because the diabetic group is older than the other groups, part of any apparent "age signature" is confounded by diabetes; this must be tested explicitly.

2 Data and preprocessing

- **Expression:** `BE700_human_expression_1.csv`, Affymetrix HG-U133 Plus 2.0 (GPL570), summarized to gene symbols and already log-normalized (no further normalization applied).
- **Phenotypes:** `BE700_human_pheno_1.csv`, the GEO series matrix sample table; the informative columns are those ending in `:ch1`.

```
expr_raw <- read.csv("data/BE700_human_expression_1.csv", check.names = FALSE,
                     stringsAsFactors = FALSE)
genes <- expr_raw[[1]]
keep  <- !is.na(genes) & genes != ""
expr  <- as.matrix(expr_raw[keep, -1]); rownames(expr) <- genes[keep]

ph_raw <- read.csv("data/BE700_human_pheno_1.csv", check.names = FALSE,
                   stringsAsFactors = FALSE)
num <- function(x) suppressWarnings(as.numeric(x))   # "missing value" -> NA
ph <- data.frame(
  sample   = ph_raw$geo_accession,
  title    = ph_raw$title,
  sex      = factor(ph_raw[["gender:ch1"]]),
  age      = num(ph_raw[["age (years):ch1"]]),
  bmi      = num(ph_raw[["body mass index (kg/m2):ch1"]]),
  whr      = num(ph_raw[["waist hip ratio:ch1"]]),
  hba1c    = num(ph_raw[["hemoglobin a1c:ch1"]]),
  fglu     = num(ph_raw[["fasting glucose (iv0gavg):ch1"]]),
  finsulin = num(ph_raw[["fasting insulin (iv0inavg):ch1"]]),
  si       = num(ph_raw[["si:ch1"]]),
  glu2h    = num(ph_raw[["2 hour glucose (ogtt):ch1"]]),
  stringsAsFactors = FALSE)
ph$group <- factor(sub("^Muscle-", "", sub("-[0-9]+$", "", ph$title)),
                   levels = c("FamilyHistoryNegative", "FamilyHistoryPositive", "Diabetes"),
                   labels = c("FH-", "FH+", "T2D"))
ph$label    <- paste0(c("FH-" = "N", "FH+" = "P", "T2D" = "D")[as.character(ph$group)],
                      sub(".*-", "", ph$title))      # short plot labels, e.g. D3, N12
ph$diabetes <- factor(ph$group == "T2D", labels = c("No", "Yes"))
rownames(ph) <- ph$sample
```

```r
stopifnot(identical(colnames(expr), ph$sample))       # QC: sample order matches
dim(expr)
```

```
## [1] 22189    50
```

The matrix has **22189 genes x 50 samples**; one row with an empty gene name was removed. Sample order matches the phenotype table.

```r
tab <- aggregate(cbind(age, bmi, hba1c, si) ~ group, data = ph,
                 FUN = function(x) round(mean(x), 1), na.action = na.pass)
tab$n      <- as.vector(table(ph$group))
tab$female <- as.vector(table(ph$group, ph$sex)[, "female"])
knitr::kable(tab[, c("group", "n", "female", "age", "bmi", "hba1c", "si")],
             caption = "Cohort by group (means; SI = insulin sensitivity).")
```

Table 1: Cohort by group (means; SI = insulin sensitivity).

group	n	female	age	bmi	hba1c	si
FH-	15	8	37.9	25.1	5.0	7.0
FH+	25	14	37.6	28.0	5.1	4.1
T2D	10	6	50.3	32.8	7.0	NA

```r
kruskal.test(age ~ group, data = ph)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  age by group
## Kruskal-Wallis chi-squared = 7.595, df = 2, p-value = 0.02243
```

```r
colSums(is.na(ph[, c("whr", "si", "glu2h")]))
```

```
##   whr    si glu2h
##     1     2    10
```

Data issues found. (i) The T2D group is older than the FH- and FH+ groups (age differs by group, Kruskal-Wallis above), so age and diabetes are confounded (H4). (ii) OGTT values are missing for all T2D subjects, SI for two, and waist-hip ratio for one (stored as the string "missing value"). (iii) One FH- subject has BMI 20 but WHR 1.36, the cohort maximum, which is probably a data-entry error; WHR is therefore not used for inference.

2.1 Quality control

```r
ord <- order(ph$group)
boxplot(expr[, ord], outline = FALSE, las = 2, cex.axis = 0.5, names = ph$label[ord],
        col = c("grey85", "lightblue", "salmon")[ph$group[ord]], ylab = "log expression")
legend("topright", fill = c("grey85", "lightblue", "salmon"), legend = levels(ph$group),
       cex = 0.7, horiz = TRUE, bty = "n")
```

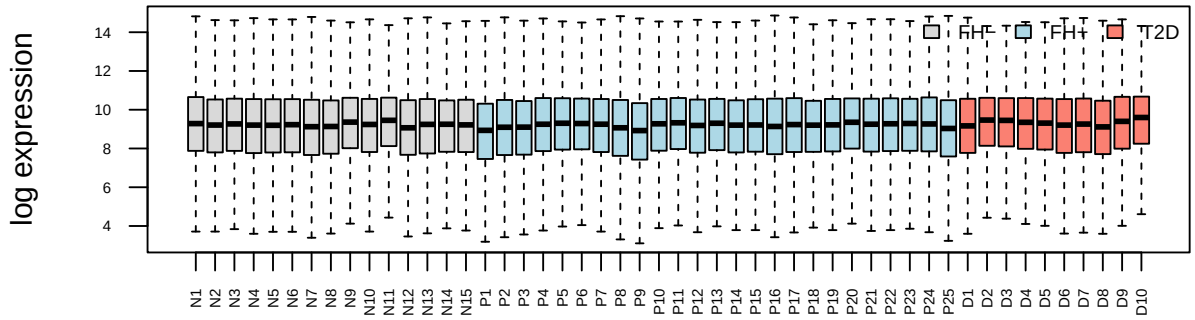


Figure 1: Per-sample expression distributions (log scale). Distributions are comparable, so no further normalization was applied.

```
sexchk <- data.frame(XIST = expr["XIST", ], RPS4Y1 = expr["RPS4Y1", ], sex = ph$sex)
ggplot(sexchk, aes(XIST, RPS4Y1, colour = sex)) + geom_point(size = 2)
```

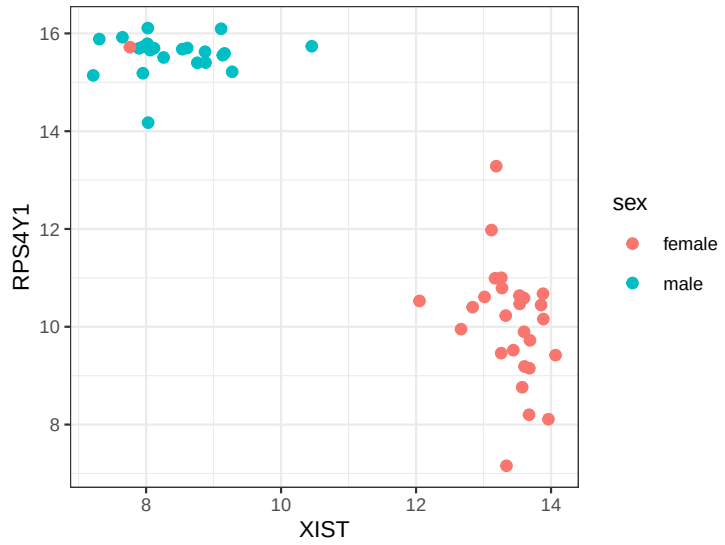


Figure 2: Sex check: XIST (X-inactivation) versus RPS4Y1 (Y-linked).

```
pred_sex <- ifelse(sexchk$XIST > 11 & sexchk$RPS4Y1 < 12, "female",
            ifelse(sexchk$XIST < 11 & sexchk$RPS4Y1 > 12, "male", "ambiguous"))
mm <- pred_sex != as.character(ph$sex)
knitr::kable(data.frame(sample = ph$label, reported = ph$sex, XIST = sexchk$XIST,
                        RPS4Y1 = sexchk$RPS4Y1, expression_says = pred_sex)[mm, ],
             digits = 1, row.names = FALSE, caption = "Samples whose expression disagrees with r
```

Table 2: Samples whose expression disagrees with reported sex.

sample	reported	XIST	RPS4Y1	expression_says
N1	female	13.2	13.3	ambiguous
P25	female	7.8	15.7	male

Reported sex agrees with XIST/RPS4Y1 expression for 48 of 50 samples. P25 is reported female but expresses Y-linked genes and lacks XIST, so it is most likely a mislabeled male. N1 expresses **both** XIST and Y-linked genes at high levels; this is consistent with a mixed / contaminated sample (or, less likely, a 47,XXY karyotype). The same subject has the implausible WHR noted above, so this record is treated with suspicion. Both samples are kept with their reported labels in the main analysis and excluded in a sensitivity analysis (Section 6).

2.2 Outlier samples

```
cc <- cor(expr)
medcor <- apply(cc, 2, function(v) median(v[v < 1]))
thr <- quantile(medcor, 0.25) - 1.5 * IQR(medcor)
outl <- ph$label[medcor < thr]
o2 <- order(medcor)
barplot(medcor[o2], names.arg = ph$label[o2], las = 2, cex.names = 0.5,
        col = ifelse(medcor[o2] < thr, "firebrick", "grey70"), ylim = c(0.75, 0.9), xpd = FALSE,
        ylab = "median r with other samples")
abline(h = thr, lty = 2)
```

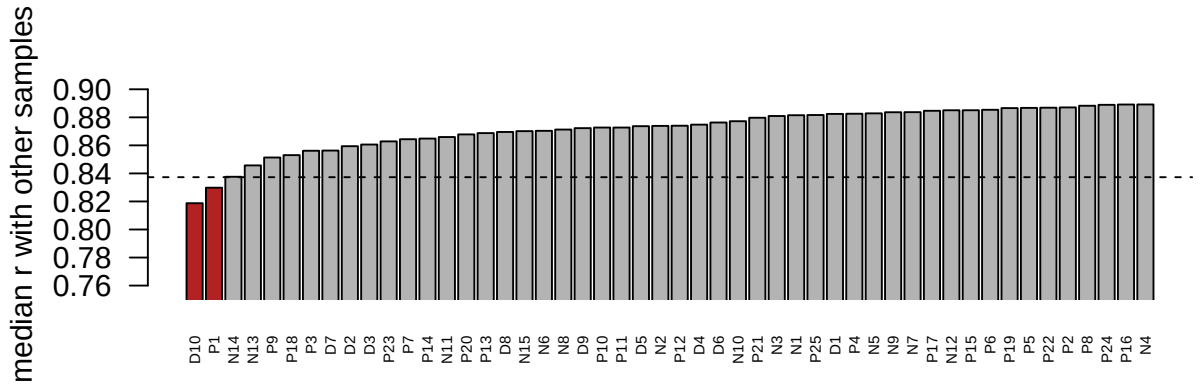


Figure 3: Median Pearson correlation of each sample with all other samples. Red: below Q1 - 1.5 IQR.

```
excl <- ph$sample[medcor < thr | mm]    # outliers + sex-discordant samples
```

The median between-sample correlation is only 0.874, so the arrays are noisy relative to typical RNA-seq data. Outlier samples: P1, D10.

3 Heatmap of the muscle transcriptome

Genes were ranked by variance across samples and the 1,000 most variable were kept; each gene was z-scored across samples. Samples and genes were clustered hierarchically (Euclidean distance, Ward.D2 linkage).

```
v     <- apply(expr, 1, var)
top   <- names(sort(v, decreasing = TRUE))[1:1000]
z     <- t(scale(t(expr[top, ])))
ann   <- data.frame(Sex = ph$sex, Group = ph$group, Age = ph$age, BMI = ph$bmi,
                    row.names = ph$sample)
```

```
ann_col <- list(Sex = c(female = "#E7298A", male = "#1B9E77"),
                Group = c("FH-" = "grey70", "FH+" = "#7570B3", "T2D" = "#D95F02"))
```

```
pheatmap(z, show_rownames = FALSE, labels_col = ph$label, fontsize_col = 6,
         annotation_col = ann, annotation_colors = ann_col,
         clustering_method = "ward.D2", treeheight_row = 0,
         breaks = seq(-3, 3, length.out = 101),
         color = colorRampPalette(c("navy", "white", "firebrick3"))(100))
```

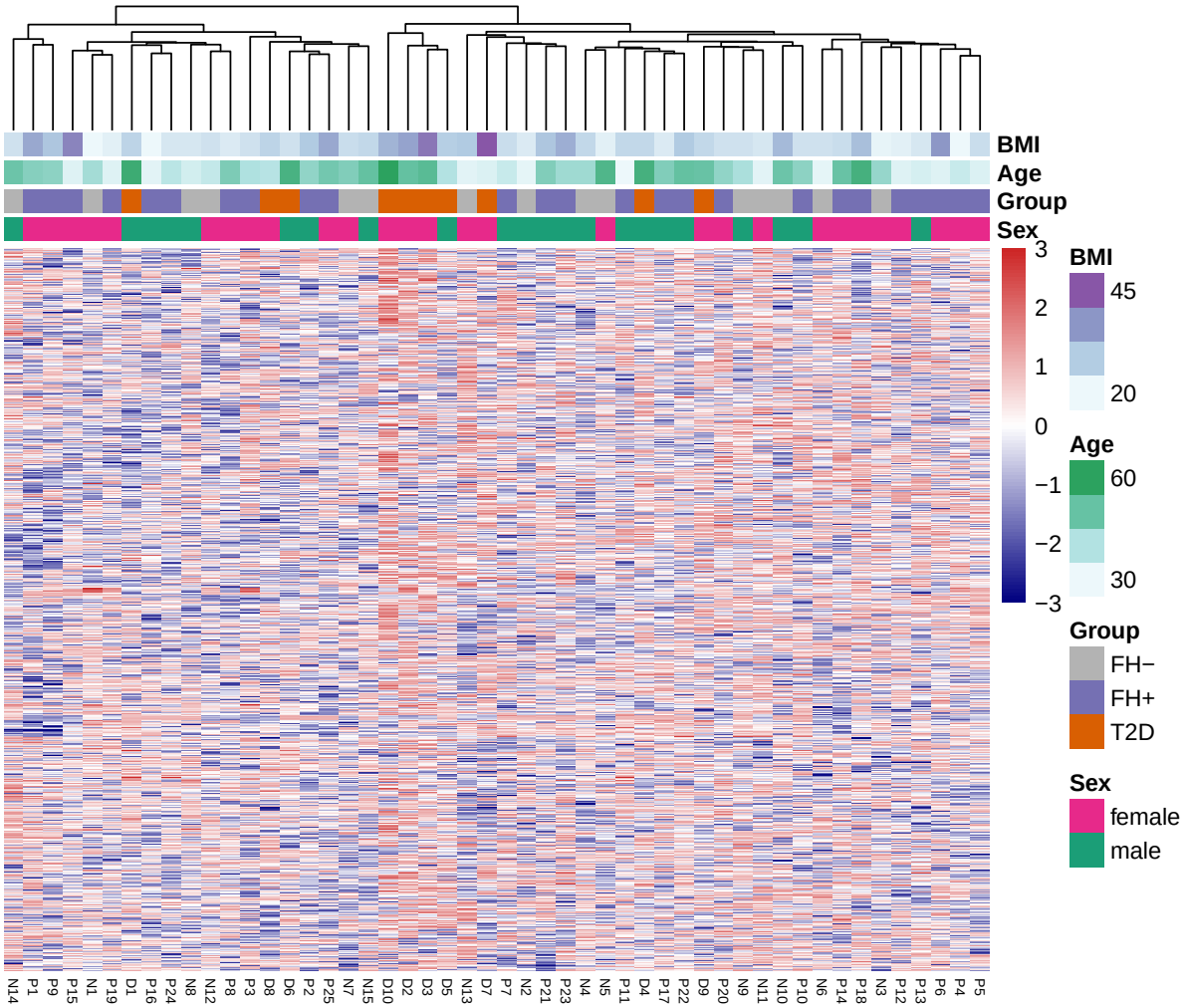


Figure 4: Top 1,000 most variable genes, z-scored, Ward.D2 clustering.

To see what lies under the sex signal, sex-associated genes were identified with a Welch t-test (BH q < 0.05 and $|\text{mean difference}| > 1$ log unit) and removed, and the heatmap was redrawn on the 1,000 most variable remaining genes.

```
tt <- apply(expr, 1, function(g) t.test(g ~ ph$sex)$p.value)
dm <- rowMeans(expr[, ph$sex == "male"]) - rowMeans(expr[, ph$sex == "female"])
sex_genes <- names(which(p.adjust(tt, "BH") < 0.05 & abs(dm) > 1))
length(sex_genes); head(sort(sex_genes), 25)
```

```
## [1] 19
```

```
##  [1] "ADH1B"        "CD24"         "CYTH2"        "DDX3Y"        "EIF1AY"
##  [6] "IRX3"         "KDM5D"        "LEP"          "LINC00844"    "LOC100506563"
## [11] "RPS4Y1"       "TPRG1"        "TSIX"         "TTTY14"       "TXLNGY"
## [16] "USP9Y"        "UTY"          "XIST"         "ZFY"
```

```r
expr_ns <- expr[setdiff(rownames(expr), sex_genes), ]
top_ns  <- names(sort(apply(expr_ns, 1, var), decreasing = TRUE))[1:1000]
z_ns    <- t(scale(t(expr_ns[top_ns, ])))
```

```r
pheatmap(z_ns, show_rownames = FALSE, labels_col = ph$label, fontsize_col = 6,
         annotation_col = ann, annotation_colors = ann_col,
         clustering_method = "ward.D2", treeheight_row = 0,
         breaks = seq(-3, 3, length.out = 101),
         color = colorRampPalette(c("navy", "white", "firebrick3"))(100))
```

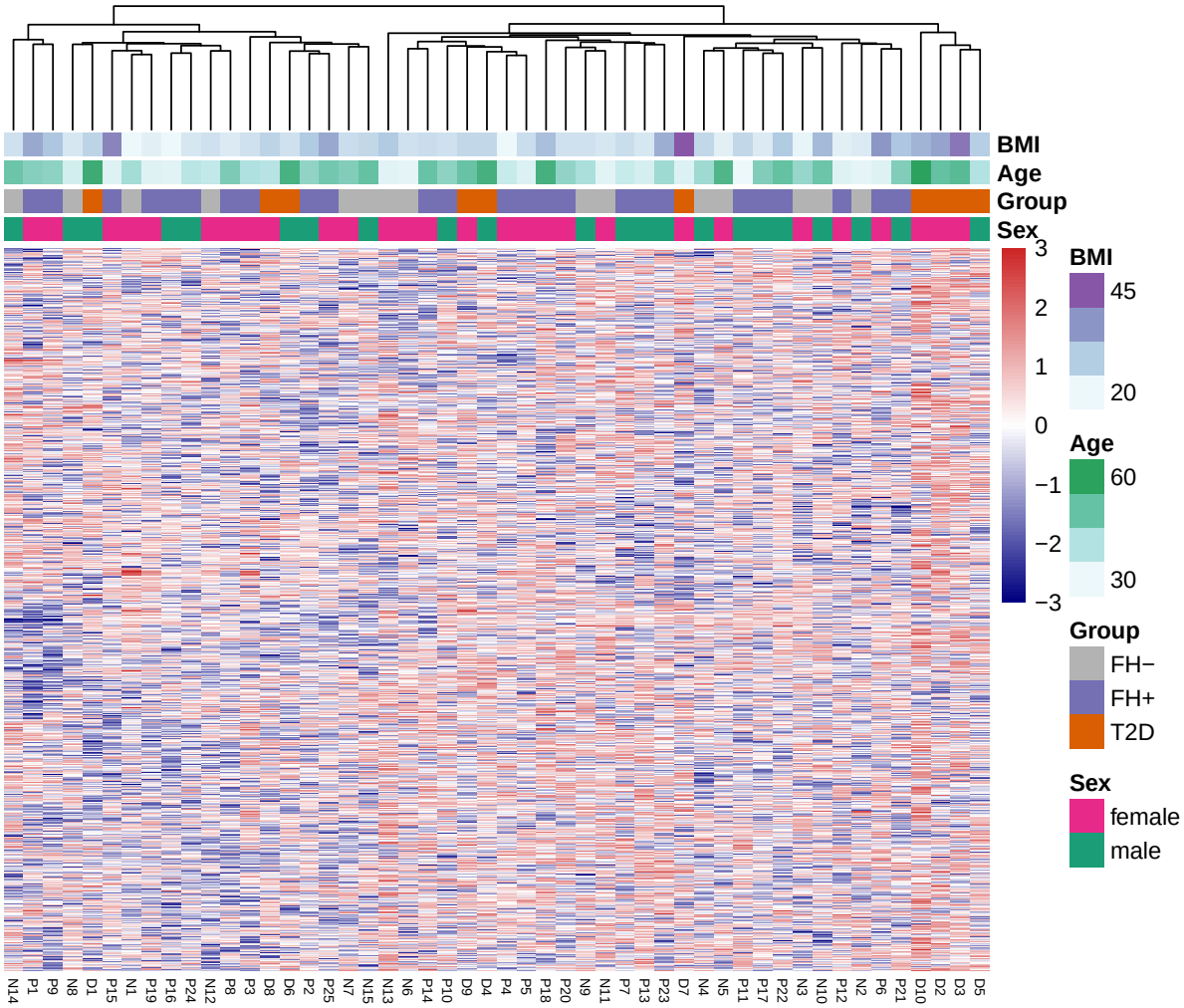


Figure 5: Top 1,000 most variable genes after removing sex-associated genes.

4 Clustering patients

Both k-means (25 random starts) and hierarchical clustering (Ward.D2) were applied to the samples, on the z-scored top-1,000 genes with and without sex-associated genes. The number of clusters was chosen by average silhouette width for k = 2 to 6, and agreement between methods was measured with the adjusted Rand index (ARI).

```
ari <- function(a, b) {                          # adjusted Rand index
  tab <- table(a, b); n <- sum(tab); c2 <- function(x) sum(choose(x, 2))
  e <- c2(rowSums(tab)) * c2(colSums(tab)) / choose(n, 2)
  (c2(tab) - e) / ((c2(rowSums(tab)) + c2(colSums(tab))) / 2 - e)
}
run_clusters <- function(zmat) {
  X  <- t(zmat); d <- dist(X); hc <- hclust(d, "ward.D2")
  sil <- sapply(2:6, function(k) {
    km <- kmeans(X, k, nstart = 25)
    c(kmeans = mean(silhouette(km$cluster, d)[, 3]),
      hclust = mean(silhouette(cutree(hc, k), d)[, 3]))
  })
  colnames(sil) <- paste0("k=", 2:6)
  list(X = X, d = d, hc = hc, sil = sil)
}
cl_all <- run_clusters(z); cl_ns <- run_clusters(z_ns)
round(cl_all$sil, 3); round(cl_ns$sil, 3)
```

```
##          k=2   k=3   k=4   k=5   k=6
## kmeans 0.021 0.019 0.018 0.014 0.015
## hclust 0.021 0.020 0.017 0.013 0.011
```

```
##          k=2   k=3   k=4   k=5   k=6
## kmeans 0.021 0.021 0.019 0.016 0.013
## hclust 0.021 0.020 0.017 0.012 0.013
```

```
k_all <- which.max(cl_all$sil["kmeans", ]) + 1
k_ns  <- which.max(cl_ns$sil["kmeans", ]) + 1
ph$km_all <- factor(kmeans(cl_all$X, k_all, nstart = 25)$cluster)
ph$hc_all <- factor(cutree(cl_all$hc, k_all))
ph$km_ns  <- factor(kmeans(cl_ns$X, k_ns, nstart = 25)$cluster)
ph$hc_ns  <- factor(cutree(cl_ns$hc, k_ns))
c(k_all = k_all, ARI_all = round(ari(ph$km_all, ph$hc_all), 2),
  k_nosex = k_ns, ARI_nosex = round(ari(ph$km_ns, ph$hc_ns), 2))
```

```
##   k_all.k=2     ARI_all k_nosex.k=2   ARI_nosex
##        2.00        0.57        2.00        0.57
```

```
table(kmeans_all = ph$km_all, sex = ph$sex)
```

```
##           sex
## kmeans_all female male
##          1     13   12
##          2     15   10
```

```r
table(kmeans_nosex = ph$km_ns, group = ph$group)
```

```
##              group
## kmeans_nosex FH- FH+ T2D
##            1   6  12   7
##            2   9  13   3
```

4.1 Are clusters enriched for phenotypes?

Categorical phenotypes were tested with Fisher's exact test and continuous phenotypes with the Kruskal-Wallis test; p-values were BH-adjusted within each clustering.

```r
enrich <- function(cl) {
  cat_v <- c(sex = "sex", group = "group", diabetes = "diabetes")
  con_v <- c(age = "age", BMI = "bmi", HbA1c = "hba1c", SI = "si",
             fasting_insulin = "finsulin", fasting_glucose = "fglu")
  p <- c(sapply(cat_v, function(v) fisher.test(table(cl, ph[[v]]))$p.value),
         sapply(con_v, function(v) kruskal.test(ph[[v]] ~ cl)$p.value))
  data.frame(phenotype = names(p), p = signif(p, 2), q_BH = signif(p.adjust(p, "BH"), 2))
}
en <- merge(setNames(enrich(ph$km_all), c("phenotype", "p_all", "q_all")),
            setNames(enrich(ph$km_ns),  c("phenotype", "p_nosex", "q_nosex")))
knitr::kable(en[order(en$q_all), ], row.names = FALSE,
             caption = "Phenotype enrichment of k-means clusters, all genes vs sex genes removed
```

Table 3: Phenotype enrichment of k-means clusters, all genes vs sex genes removed.

phenotype	p_all	q_all	p_nosex	q_nosex
BMI	0.073	0.22	0.073	0.22
fasting_insulin	0.067	0.22	0.067	0.22
SI	0.033	0.22	0.033	0.22
diabetes	0.290	0.65	0.290	0.65
group	0.380	0.68	0.380	0.68
age	0.850	0.85	0.850	0.85
fasting_glucose	0.830	0.85	0.830	0.85
HbA1c	0.740	0.85	0.740	0.85
sex	0.780	0.85	0.780	0.85

```r
write.csv(en, "results/cluster_enrichment.csv", row.names = FALSE)
```

```r
long <- do.call(rbind, lapply(c("age", "bmi", "hba1c", "si"), function(v)
  data.frame(cluster = ph$km_ns, variable = v, value = ph[[v]])))
ggplot(long, aes(cluster, value, fill = cluster)) + geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.15, size = 0.8) + facet_wrap(~variable, scales = "free_y", nrow = 1) +
  theme(legend.position = "none")
```

Do male and female patients form distinct clusters? With all genes, the k-means partition (k = 2) against sex has Fisher p = 0.78. After sex-associated genes are removed, that association becomes p

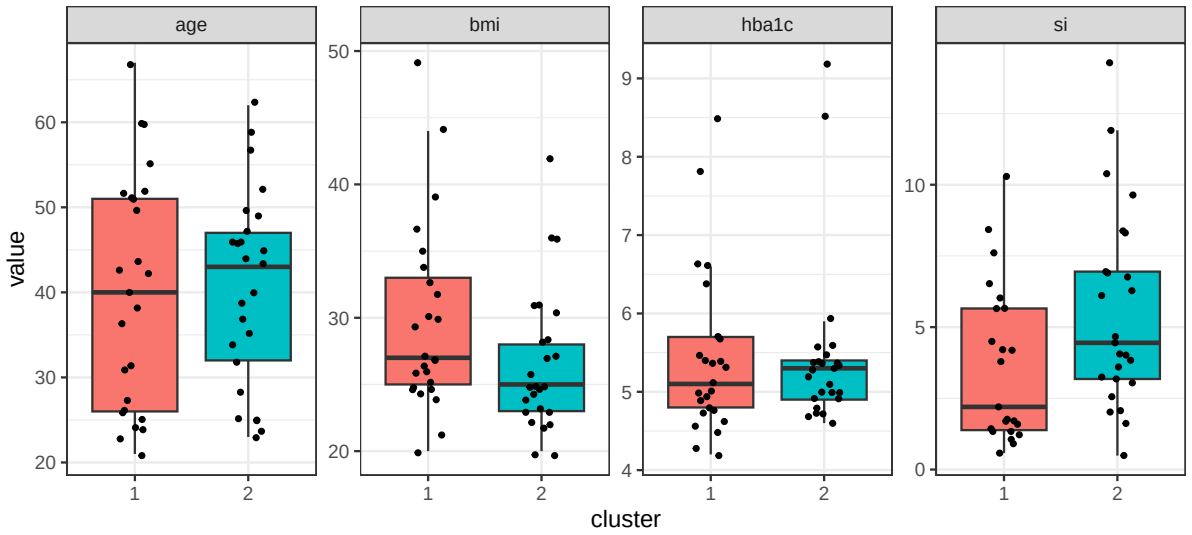


Figure 6: Phenotypes by k-means cluster (sex-associated genes removed).

$= 0.78$.

5 Genes correlated with age

For every gene, the Pearson and Spearman correlation with age was computed across all 50 subjects, with Benjamini-Hochberg FDR. Because the T2D group is older (H4), two robustness checks were added: (a) a linear model `expression ~ age + sex + diabetes` fitted to every gene, and (b) the correlation re-computed in the 40 non-diabetic subjects only.

```
cor_test_vec <- function(M, x, method = "pearson") {
  r <- as.vector(cor(t(M), x, method = method, use = "pairwise.complete.obs"))
  n <- sum(!is.na(x)); tstat <- r * sqrt((n - 2) / (1 - r^2))
  p <- 2 * pt(-abs(tstat), n - 2)
  data.frame(gene = rownames(M), r = r, p = p, q = p.adjust(p, "BH"))
}
pear <- cor_test_vec(expr, ph$age, "pearson")
spear <- cor_test_vec(expr, ph$age, "spearman")
nd <- ph$diabetes == "No"
pear_nd <- cor_test_vec(expr[, nd], ph$age[nd], "pearson")

# Vectorized OLS: expression ~ age + sex + diabetes, t-test on the age coefficient
X <- model.matrix(~ age + sex + diabetes, ph)
B <- solve(crossprod(X), crossprod(X, t(expr)))
res <- t(expr) - X %*% B
s2 <- colSums(res^2) / (nrow(X) - ncol(X))
se_age <- sqrt(s2 * solve(crossprod(X))["age", "age"])
t_age <- B["age", ] / se_age
p_lm <- 2 * pt(-abs(t_age), nrow(X) - ncol(X))

age_tab <- data.frame(gene = pear$gene, r_pearson = pear$r, q_pearson = pear$q,
                      r_spearman = spear$r, q_spearman = spear$q,
```

```
                      r_nondiabetic = pear_nd$r, p_nondiabetic = pear_nd$p,
                      beta_age_adj = B["age", ], q_adj = p.adjust(p_lm, "BH"))
age_tab <- age_tab[order(pear$p), ]
write.csv(age_tab, "results/age_correlation_all_genes.csv", row.names = FALSE)

sig <- c(pearson_q05 = sum(age_tab$q_pearson < 0.05),
         spearman_q05 = sum(age_tab$q_spearman < 0.05),
         adjusted_q05 = sum(age_tab$q_adj < 0.05),
         nondiabetic_p001 = sum(age_tab$p_nondiabetic < 0.001),
         pearson_p001 = sum(pear$p < 0.001))
sig
```

```
##      pearson_q05     spearman_q05     adjusted_q05 nondiabetic_p001
##                0                1                1               35
##     pearson_p001
##               53
```

```
# Age-gene set: Pearson FDR < 0.05; fall back to the top 100 by p if fewer than 30 pass
age_genes <- if (sig["pearson_q05"] >= 30) age_tab$gene[age_tab$q_pearson < 0.05] else
  head(age_tab$gene, 100)
age_rule  <- if (sig["pearson_q05"] >= 30) "Pearson FDR < 0.05" else "top 100 by Pearson p"
robust    <- age_tab$gene[age_tab$gene %in% age_genes & age_tab$q_adj < 0.1 &
                          sign(age_tab$r_nondiabetic) == sign(age_tab$r_pearson)]
c(n_age_genes = length(age_genes), n_up = sum(age_tab$r_pearson[age_tab$gene %in% age_genes] > 0
  n_robust = length(robust))
```

```
## n_age_genes        n_up    n_robust
##         100          44           1
```

```
knitr::kable(head(age_tab[, c("gene", "r_pearson", "q_pearson", "r_spearman",
                              "r_nondiabetic", "q_adj")], 20),
             digits = 3, row.names = FALSE,
             caption = "Top 20 age-correlated genes. q_adj: FDR for age in expression ~ age + se
```

Table 4: Top 20 age-correlated genes. q_adj: FDR for age in expression ~ age + sex + diabetes.

gene	r_pearson	q_pearson	r_spearman	r_nondiabetic	q_adj
UNC13C	0.612	0.051	0.705	0.638	0.043
MFF	-0.597	0.052	-0.591	-0.462	0.256
SCN4B	-0.578	0.083	-0.582	-0.464	0.266
SLIT2	0.544	0.188	0.549	0.417	0.419
PCDH9	0.541	0.188	0.517	0.469	0.203
C2orf88	-0.540	0.188	-0.513	-0.379	0.431
MRPL15	-0.534	0.203	-0.542	-0.347	0.401
DLAT	-0.525	0.210	-0.435	-0.342	0.466
RAVER2	-0.525	0.210	-0.516	-0.697	0.308
FAM96A	-0.523	0.210	-0.520	-0.407	0.438
AP2B1	-0.519	0.210	-0.534	-0.499	0.301

gene	r_pearson	q_pearson	r_spearman	r_nondiabetic	q_adj
CADM2	-0.518	0.210	-0.516	-0.335	0.438
DCTPP1	-0.514	0.210	-0.479	-0.271	0.495
MDH2	-0.514	0.210	-0.537	-0.525	0.298
STRADB	-0.511	0.210	-0.493	-0.300	0.377
DSCAM-AS1	0.510	0.210	0.519	0.410	0.441
PTRHD1	-0.509	0.210	-0.544	-0.378	0.298
COMMD7	0.505	0.212	0.485	0.440	0.298
THNSL1	-0.504	0.212	-0.482	-0.464	0.431
ATP5F1	-0.504	0.212	-0.502	-0.403	0.438

The age-gene set (top 100 by Pearson p) contains **100 genes**; 1 of them stay significant after adjusting for sex and diabetes ($q < 0.1$) and keep the same direction in non-diabetic subjects.

5.1 Heatmap of age-correlated genes

```
oa <- order(ph$age)
za <- t(scale(t(expr[age_genes, oa])))
pheatmap(za, cluster_cols = FALSE, show_rownames = length(age_genes) <= 60,
         fontsize_row = 5, labels_col = paste0(ph$label[oa], " (", ph$age[oa], ")"),
         fontsize_col = 5, annotation_col = ann, annotation_colors = ann_col,
         clustering_method = "ward.D2", treeheight_row = 20,
         breaks = seq(-3, 3, length.out = 101),
         color = colorRampPalette(c("navy", "white", "firebrick3"))(100))
```

5.2 Correlation plots for the top age-correlated genes

```
top9 <- head(age_tab$gene, 9)
sc <- do.call(rbind, lapply(top9, function(g) data.frame(
  gene = sprintf("%s (r = %.2f)", g, age_tab$r_pearson[age_tab$gene == g]),
  age = ph$age, expr = expr[g, ], group = ph$group)))
sc$gene <- factor(sc$gene, levels = unique(sc$gene))
ggplot(sc, aes(age, expr)) + geom_point(aes(colour = group), size = 1.4) +
  geom_smooth(method = "lm", se = TRUE, colour = "black", linewidth = 0.6) +
  scale_colour_manual(values = ann_col$Group) +
  facet_wrap(~gene, scales = "free_y") + labs(x = "Age (years)", y = "log expression")
```

```
t30 <- head(age_tab, 30)
bar <- rbind(data.frame(gene = t30$gene, r = t30$r_pearson, set = "all (n = 50)"),
             data.frame(gene = t30$gene, r = t30$r_nondiabetic, set = "non-diabetic (n = 40)"))
bar$gene <- factor(bar$gene, levels = rev(t30$gene))
ggplot(bar, aes(r, gene, fill = set)) + geom_col(position = "dodge", width = 0.75) +
  geom_vline(xintercept = 0) + labs(x = "Pearson r with age", y = NULL) +
  theme(axis.text.y = element_text(size = 6), legend.position = "top")
```

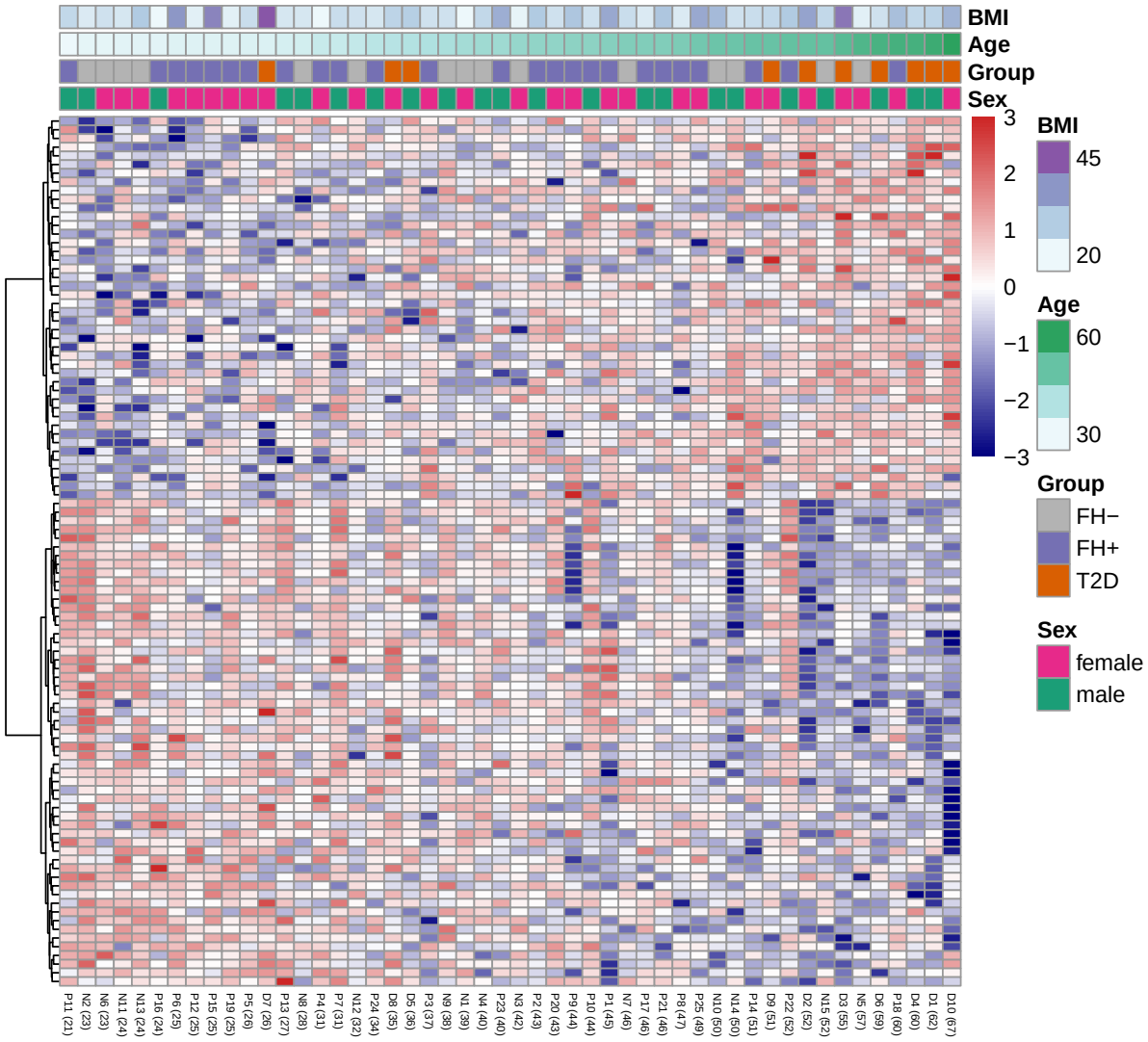


Figure 7: Age-correlated genes; samples ordered by age (left = youngest). Genes clustered.

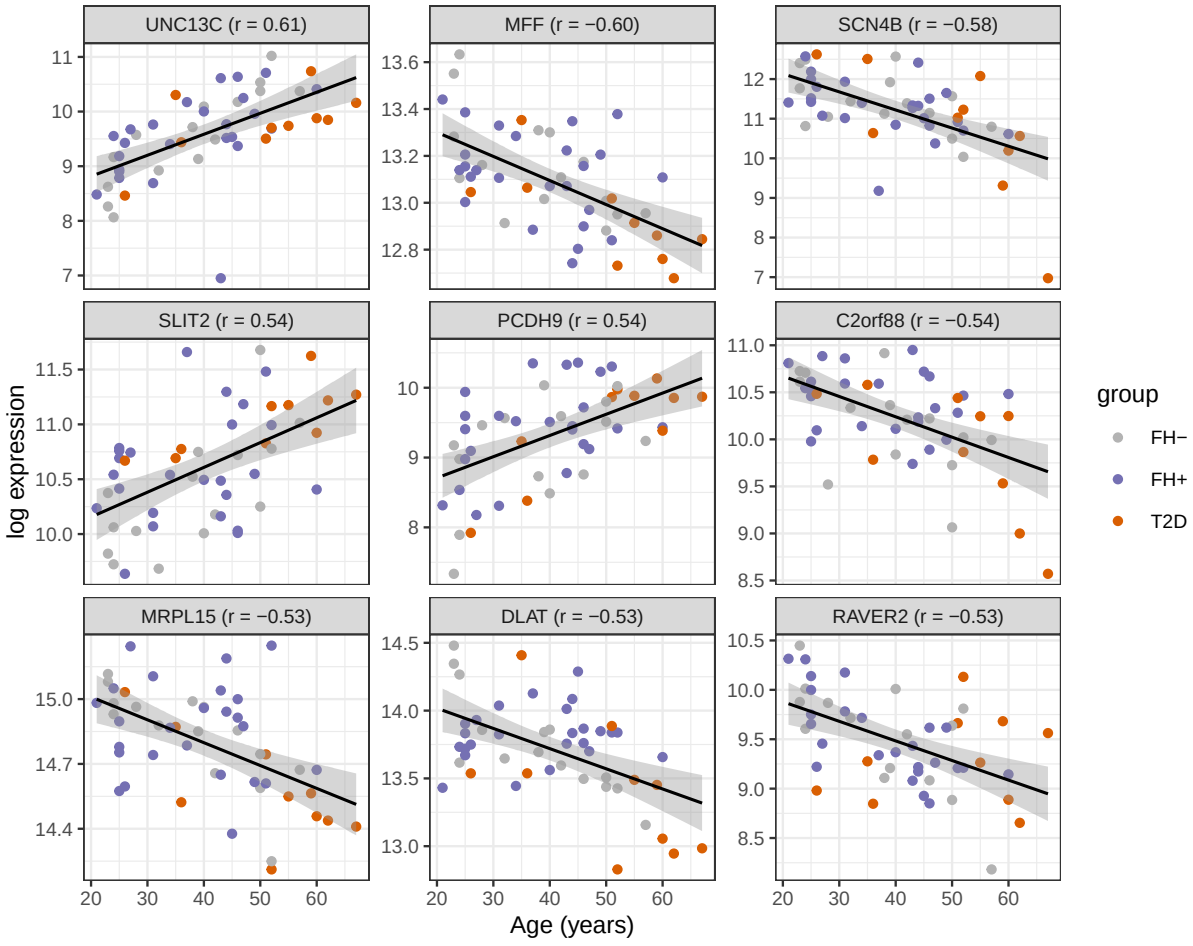


Figure 8: Top 9 age-correlated genes; T2D subjects in orange. Lines: linear fit on all subjects.

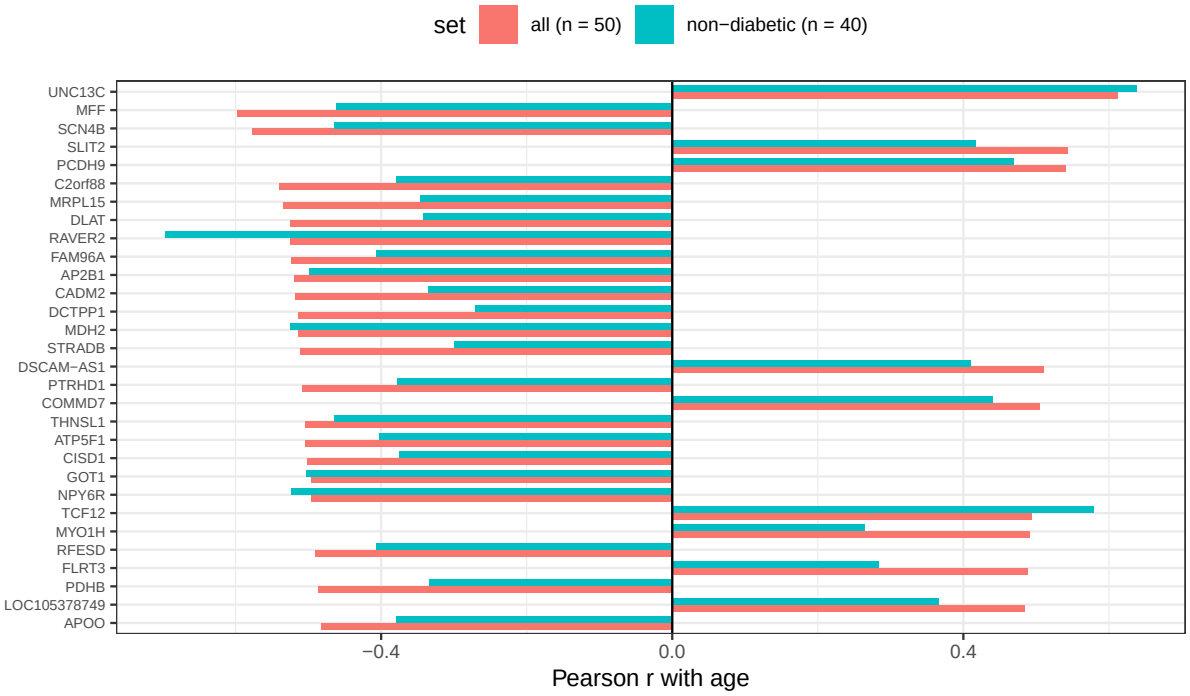


Figure 9: Top 30 age-correlated genes: Pearson r, all subjects vs non-diabetic subjects only.

6 Principal component analysis

PCA (`prcomp`) was run on all 22189 genes, each centered to mean zero but not scaled (the data are already on a log scale, so unscaled PCA keeps high-variance genes influential). It was repeated after removing sex-associated genes. Each point is labeled with the subject code (N = FH-, P = FH+, D = T2D, followed by the GEO sample number within the group).

```
pca    <- prcomp(t(expr), center = TRUE, scale. = FALSE)
pca_ns <- prcomp(t(expr_ns), center = TRUE, scale. = FALSE)
ve     <- 100 * pca$sdev^2 / sum(pca$sdev^2)
ve_ns  <- 100 * pca_ns$sdev^2 / sum(pca_ns$sdev^2)
round(rbind(all = ve[1:6], no_sex = ve_ns[1:6]), 1)
```

```
##        [,1] [,2] [,3] [,4] [,5] [,6]
## all     7.6  4.4  3.9  3.5    3  2.8
## no_sex  7.6  4.4  3.9  3.5    3  2.8
```

```
pcd <- cbind(ph, pca$x[, 1:6]); pcd_ns <- cbind(ph, pca_ns$x[, 1:6])
```

```
ggplot(pcd, aes(PC1, PC2, colour = sex, shape = group)) + geom_point(size = 2.3) +
  geom_text_repel(aes(label = label), size = 2.4, max.overlaps = 50, show.legend = FALSE) +
  scale_colour_manual(values = ann_col$Sex) +
  labs(x = sprintf("PC1 (%.1f%%)", ve[1]), y = sprintf("PC2 (%.1f%%)", ve[2]))
```

```
ggplot(pcd_ns, aes(PC1, PC2, colour = age, shape = group)) + geom_point(size = 2.3) +
  geom_text_repel(aes(label = label), size = 2.4, max.overlaps = 50, show.legend = FALSE) +
  scale_colour_viridis_c() +
  labs(x = sprintf("PC1 (%.1f%%)", ve_ns[1]), y = sprintf("PC2 (%.1f%%)", ve_ns[2]))
```

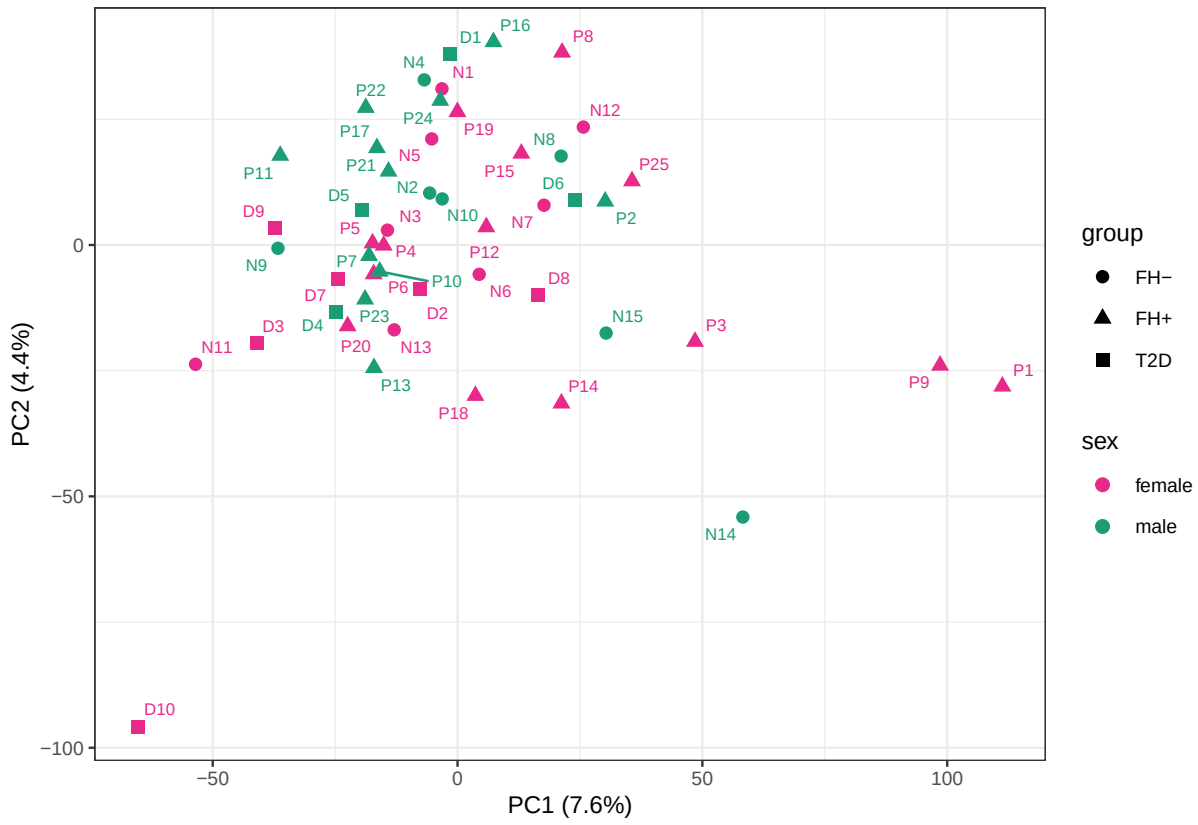


Figure 10: PCA of all genes, colored by sex, shaped by group.

Sensitivity analysis. The first PCs are pulled by a few individual samples (e.g. D10 alone drives PC2). PCA was therefore repeated without the outlier and sex-discordant samples (N1, P1, D10, P25), using the 1,000 most variable non-sex genes, each scaled to unit variance so no single gene dominates.

```
keep_s  <- !(ph$sample %in% excl)
pca_s   <- prcomp(t(expr_ns[top_ns, keep_s]), center = TRUE, scale. = TRUE)
ve_s    <- 100 * pca_s$sdev^2 / sum(pca_s$sdev^2)
pcd_s   <- cbind(ph[keep_s, ], pca_s$x[, 1:6])
ggplot(pcd_s, aes(PC1, PC2, colour = group, size = bmi)) + geom_point(alpha = 0.85) +
  geom_text_repel(aes(label = label), size = 2.4, max.overlaps = 50, show.legend = FALSE) +
  scale_colour_manual(values = ann_col$Group) + scale_size(range = c(1.2, 4)) +
  labs(x = sprintf("PC1 (%.1f%%)", ve_s[1]), y = sprintf("PC2 (%.1f%%)", ve_s[2]))
```

```
pc_assoc <- function(d) {
  vars <- c("age", "bmi", "hba1c", "si", "finsulin")
  out <- sapply(paste0("PC", 1:6), function(pc) c(
    sex = wilcox.test(d[[pc]] ~ d$sex)$p.value,
    group = kruskal.test(d[[pc]] ~ d$group)$p.value,
    sapply(vars, function(v) cor.test(d[[pc]], d[[v]], method = "spearman", exact = FALSE)$p.val
  signif(out, 2)
}
knitr::kable(pc_assoc(pcd), caption = "P-values: PCs (all genes) vs phenotypes (Wilcoxon / Krusk
```

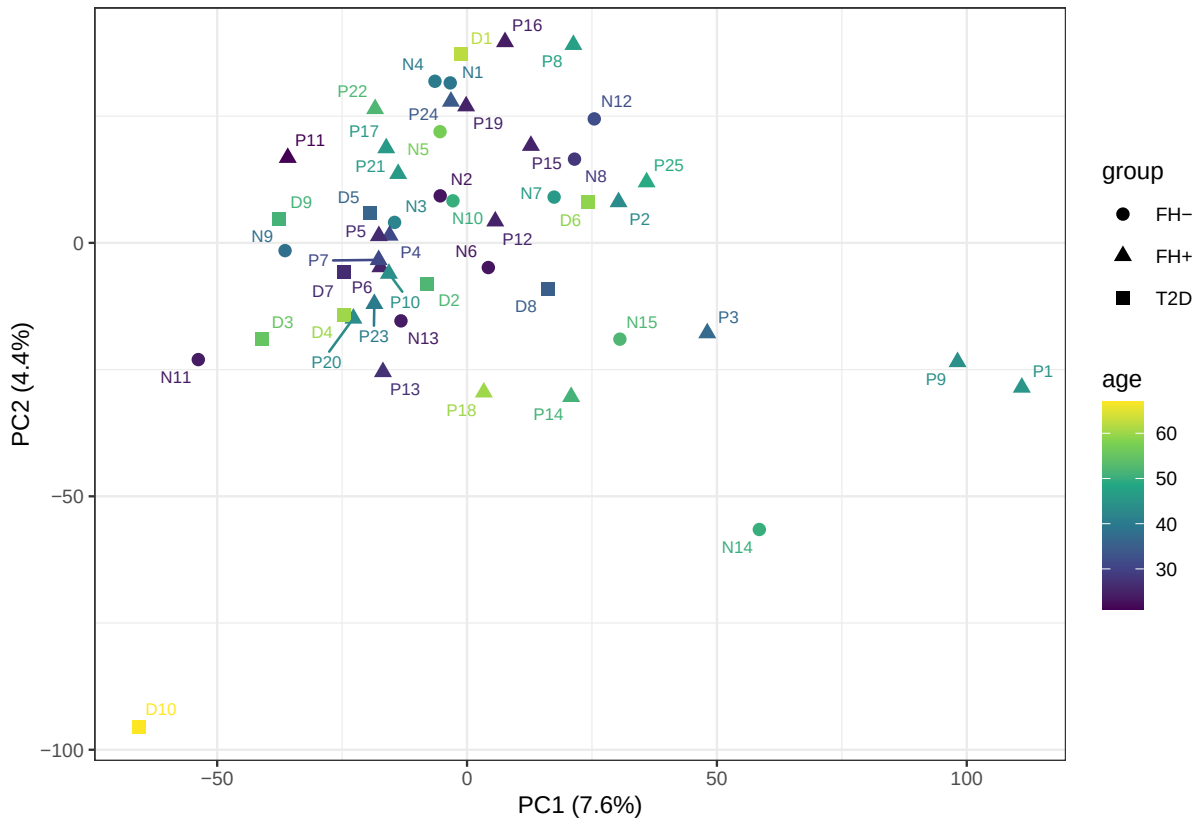


Figure 11: PCA after removing sex-associated genes, colored by age, shaped by group.

Table 5: P-values: PCs (all genes) vs phenotypes (Wilcoxon / Kruskal-Wallis / Spearman).

	PC1	PC2	PC3	PC4	PC5	PC6
sex	0.480	0.059	0.33	0.75	0.220	0.030
group	0.090	0.550	0.36	1.00	0.330	0.071
age	0.580	0.300	0.19	0.48	0.049	0.820
bmi	0.500	0.017	0.72	0.53	0.510	0.260
hba1c	0.890	0.970	0.58	0.87	0.032	0.170
si	0.078	0.730	0.11	0.27	0.800	0.150
finsulin	0.085	0.980	0.17	0.52	0.570	0.035

```
knitr::kable(pc_assoc(pcd_ns), caption = "P-values: PCs (sex-associated genes removed) vs phenot
```

Table 6: P-values: PCs (sex-associated genes removed) vs phenotypes.

	PC1	PC2	PC3	PC4	PC5	PC6
sex	0.530	0.087	0.360	0.79	0.330	0.200
group	0.089	0.540	0.380	1.00	0.350	0.045
age	0.620	0.260	0.210	0.49	0.042	0.430

	PC1	PC2	PC3	PC4	PC5	PC6
bmi	0.500	0.019	0.750	0.51	0.480	0.130
hba1c	0.860	0.990	0.640	0.87	0.038	0.100
si	0.073	0.690	0.094	0.28	0.820	0.140
finsulin	0.083	0.990	0.160	0.50	0.630	0.028

```
knitr::kable(pc_assoc(pcd_s), caption = "P-values: sensitivity PCA (outliers removed) vs phenoty
```

Table 7: P-values: sensitivity PCA (outliers removed) vs phenotypes.

	PC1	PC2	PC3	PC4	PC5	PC6
sex	0.79	0.320	0.015	0.06	0.300	0.960
group	0.49	0.480	0.810	0.27	0.510	0.019
age	0.59	0.630	0.420	0.20	0.190	0.320
bmi	0.13	0.770	0.320	0.23	0.400	0.270
hba1c	0.46	0.760	0.640	0.17	0.300	0.073
si	0.46	0.018	0.740	0.19	0.260	0.019
finsulin	0.60	0.031	0.810	0.53	0.098	0.076

```
ld <- pca_ns$rotation
data.frame(PC1_top_pos = names(head(sort(ld[, 1], decreasing = TRUE), 10)),
           PC1_top_neg = names(head(sort(ld[, 1]), 10)),
           PC2_top_pos = names(head(sort(ld[, 2], decreasing = TRUE), 10)),
           PC2_top_neg = names(head(sort(ld[, 2]), 10)))
```

```
##        PC1_top_pos       PC1_top_neg             PC2_top_pos  PC2_top_neg
## 1            RPS11              PFN1                   CRIP2 LOC101928865
## 2           ADIPOQ             HSPB6                     DDN      C7orf57
## 3           FRG1HP            COX6A2                   ASCC2        DDX53
## 4            PLAG1              SELM                   BRAT1 LOC100996325
## 5  CCT6P1///CCT6P3            IFITM3                   UBE2M        RPS11
## 6            PGBD1       HBA2///HBA1                   HCP5      SLCO1B3
## 7          SLC2A10             HSPB1                 TMEM158     ANKRD30B
## 8              GZMM LAMA5-AS1///LAMA5       SNORA5B///TBRG4        TNIP3
## 9        LOC284926         LINC00116                 PPFIA4       B3GNT7
## 10         CCDC110             UBE2M LOC101060835///HLA-DQB1     CYP4F62P
```

7 Additional analysis: insulin sensitivity and diabetes

As an exploratory extension, every gene was correlated (Spearman) with insulin sensitivity (SI, n = 48), and T2D was compared with FH- subjects (Welch t-test). Genes from Jin et al.'s SRF story (SRF, its co-activator MKL1/MRTF-A, and the SRF targets ACTA2, FHL1, VCL, CNN2) are checked individually.

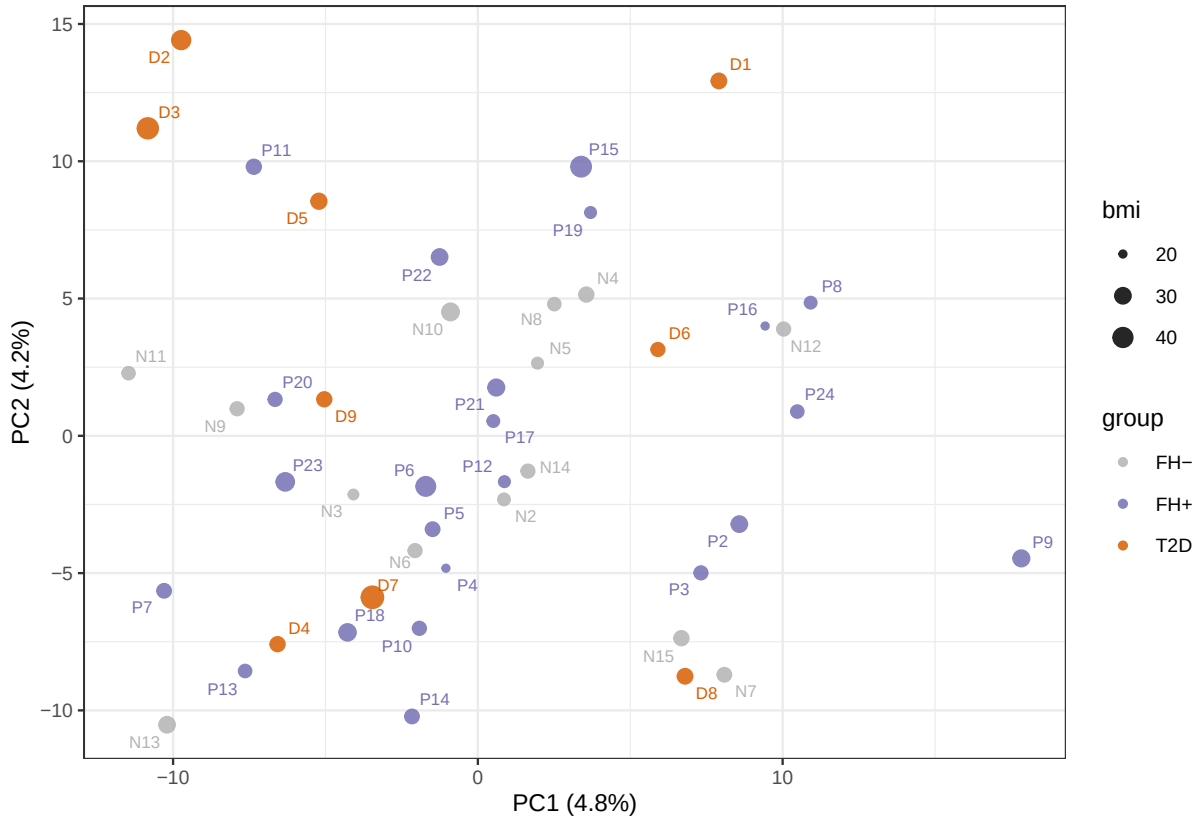


Figure 12: Sensitivity PCA: outlier and sex-discordant samples removed, top 1,000 variable non-sex genes, scaled. Colored by group, size by BMI.

```r
si_ok <- !is.na(ph$si)
si_cor <- cor_test_vec(expr[, si_ok], ph$si[si_ok], "spearman")
si_cor <- si_cor[order(si_cor$p), ]
c(si_q05 = sum(si_cor$q < 0.05), si_p001 = sum(si_cor$p < 0.001))
```

```
##  si_q05 si_p001
##       0      50
```

```r
write.csv(si_cor, "results/si_correlation_all_genes.csv", row.names = FALSE)

d_vs_n <- ph$group %in% c("T2D", "FH-")
grp <- droplevels(ph$group[d_vs_n])
de_p <- apply(expr[, d_vs_n], 1, function(g) t.test(g ~ grp)$p.value)
de_fc <- rowMeans(expr[, ph$group == "T2D"]) - rowMeans(expr[, ph$group == "FH-"])
de <- data.frame(gene = names(de_p), log_diff_T2D_vs_FHneg = de_fc, p = de_p,
                 q = p.adjust(de_p, "BH"))[order(de_p), ]
c(de_q05 = sum(de$q < 0.05), de_p001 = sum(de$p < 0.001))
```

```
##  de_q05 de_p001
##       0      44
```

```r
write.csv(de, "results/T2D_vs_FHneg.csv", row.names = FALSE)

srf <- intersect(c("SRF", "MKL1", "MRTFA", "ACTA2", "FHL1", "VCL", "CNN2", "MYH11"), rownames(ex
knitr::kable(data.frame(gene = srf,
  r_SI = si_cor$r[match(srf, si_cor$gene)], p_SI = si_cor$p[match(srf, si_cor$gene)],
  diff_T2D_FHneg = de$log_diff_T2D_vs_FHneg[match(srf, de$gene)],
  p_T2D_FHneg = de$p[match(srf, de$gene)]), digits = 3, row.names = FALSE,
  caption = "SRF pathway genes: correlation with SI and T2D vs FH- difference.")
```

Table 8: SRF pathway genes: correlation with SI and T2D vs FH- difference.

gene	r_SI	p_SI	diff_T2D_FHneg	p_T2D_FHneg
SRF	-0.370	0.010	0.294	0.017
MKL1	-0.440	0.002	0.483	0.143
ACTA2	-0.233	0.111	0.195	0.199
FHL1	-0.234	0.109	0.420	0.070
VCL	-0.070	0.638	0.585	0.041
CNN2	-0.188	0.202	0.523	0.038
MYH11	-0.053	0.720	0.095	0.802

```r
c(overlap_age_SI = length(intersect(age_genes, head(si_cor$gene, length(age_genes)))))
```

```
## overlap_age_SI
##              1
```

```r
de$lab <- ifelse(rank(de$p) <= 15, de$gene, "")
ggplot(de, aes(log_diff_T2D_vs_FHneg, -log10(p))) +
  geom_point(size = 0.4, alpha = 0.4, colour = "grey40") +
```

```
  geom_point(data = de[de$lab != "", ], colour = "#D95F02", size = 1.2) +
  geom_text_repel(aes(label = lab), size = 2.5, max.overlaps = 30) +
  labs(x = "mean log expression, T2D minus FH-", y = "-log10 p")
```

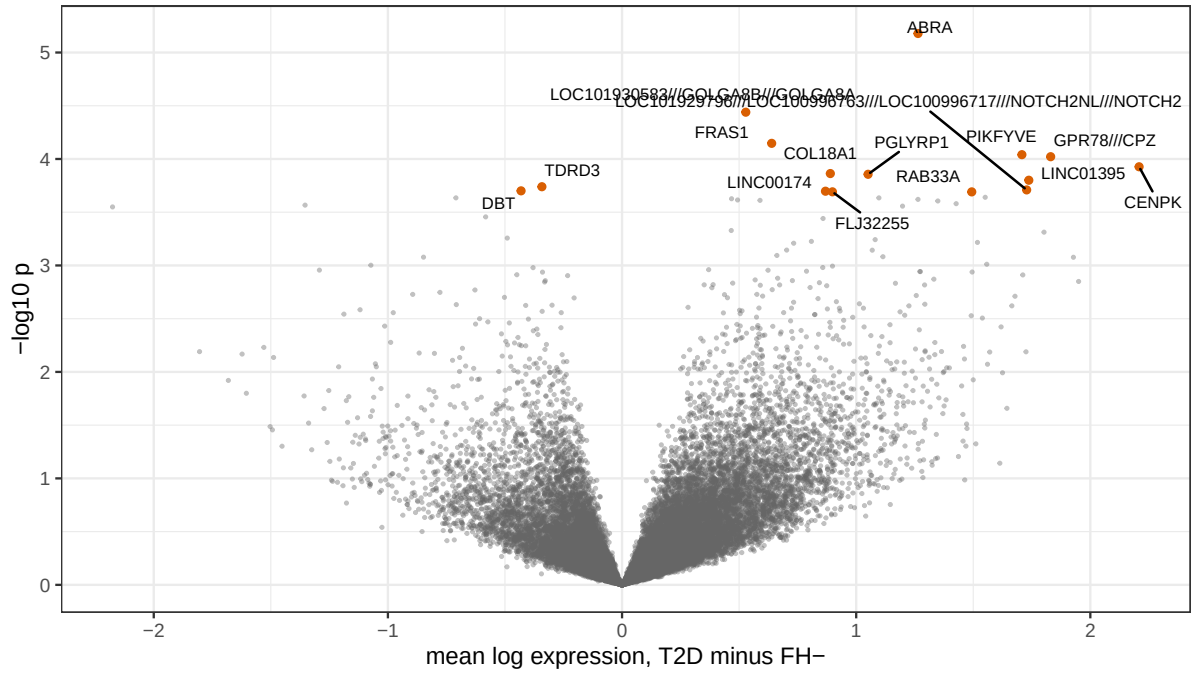


Figure 13: T2D vs FH-: difference in mean log expression vs significance.

8 DAVID functional enrichment (Part III)

Five gene lists are written to `results/` and analyzed with the DAVID Functional Annotation Chart (https://davidbioinformatics.nih.gov, run on October 21, 2026; the site listed Knowledgebase v2026_2 as its latest release): genes decreasing with age (`age_down`), genes increasing with age (`age_up`), the top 100 marker genes of each k-means cluster after sex-gene removal (`cluster1_up`, `cluster2_up`), and the 200 genes most correlated with insulin sensitivity (`SI_top200`). The single robust age gene was not submitted. Settings: identifier OFFICIAL_GENE_SYMBOL, species *Homo sapiens*, background = DAVID's built-in *Human Genome U133 Plus 2 Array* population (the array used here, so the background is the genes measurable on this platform), categories KEGG_PATHWAY and GOTERM_BP_DIRECT, default chart thresholds (Count >= 2, EASE p <= 0.1). Probe sets annotated with several symbols (`A///B`) were split into separate symbols before upload. DAVID is a web tool, so its output is stored unchanged in `data/david/` and read back here; FDR is DAVID's Benjamini-Hochberg-type FDR column.

```
up   <- age_tab$gene[age_tab$gene %in% age_genes & age_tab$r_pearson > 0]
down <- age_tab$gene[age_tab$gene %in% age_genes & age_tab$r_pearson < 0]
writeLines(up,   "results/DAVID_age_up.txt")
writeLines(down, "results/DAVID_age_down.txt")
writeLines(robust, "results/DAVID_age_robust.txt")
writeLines(head(si_cor$gene, 200), "results/DAVID_SI_top200.txt")
writeLines(rownames(expr), "results/DAVID_background.txt")
# cluster marker genes: top 100 genes by t-statistic for each no-sex k-means cluster
```

```r
for (k in levels(ph$km_ns)) {
  tk <- apply(expr_ns, 1, function(g) t.test(g[ph$km_ns == k], g[ph$km_ns != k])$statistic)
  writeLines(names(head(sort(tk, decreasing = TRUE), 100)),
             sprintf("results/DAVID_cluster%s_up.txt", k))
}
c(age_up = length(up), age_down = length(down), robust = length(robust))
```

```
##   age_up age_down   robust
##       44       56        1
```

```r
dv <- read.delim("data/david/DAVID_chart_results.tsv", check.names = FALSE)
dinfo <- read.delim("data/david/DAVID_run_info.tsv")
lists <- dinfo$List
dv$Category <- ifelse(dv$Category == "KEGG_PATHWAY", "KEGG", "GO-BP")
dv$Term <- sub("^(hsa[0-9]+:|GO:[0-9]+~)", "", dv$Term)
dinfo$terms_p_le_0.1 <- as.vector(table(factor(dv$List, lists)))
dinfo$terms_FDR_lt_0.05 <- as.vector(table(factor(dv$List[dv$FDR < 0.05], lists)))
knitr::kable(dinfo[, c("List", "input_symbols", "DAVID_IDs", "terms_p_le_0.1", "terms_FDR_lt_0.0
             row.names = FALSE,
             caption = "DAVID input lists: symbols submitted, symbols mapped to DAVID IDs, and K
```

Table 9: DAVID input lists: symbols submitted, symbols mapped to DAVID IDs, and KEGG + GO-BP terms returned.

List	input_symbols	DAVID_IDs	terms_p_le_0.1	terms_FDR_lt_0.05
age_down	58	47	22	8
age_up	47	42	6	0
cluster1_up	106	92	18	0
cluster2_up	109	102	19	0
SI_top200	246	211	18	0

```r
top <- do.call(rbind, lapply(split(dv, list(dv$List, dv$Category), drop = TRUE),
                             function(d) head(d[order(d$PValue), ], 3)))
top <- top[order(match(top$List, lists), top$Category != "KEGG", top$PValue), ]
top$Term <- ifelse(nchar(top$Term) > 48, paste0(substr(top$Term, 1, 45), "..."), top$Term)
top$fold <- round(top$`Fold Enrichment`, 1)
top$p <- signif(top$PValue, 2); top$FDR <- signif(top$FDR, 2)
knitr::kable(top[, c("List", "Category", "Term", "Count", "fold", "p", "FDR")], row.names = FALS
             caption = "Top 3 KEGG and top 3 GO-BP terms per list (DAVID EASE p; FDR < 0.05 only
```

Table 10: Top 3 KEGG and top 3 GO-BP terms per list (DAVID EASE p; FDR < 0.05 only in age_down).

List	Category	Term	Count	fold	p	FDR
age_down	KEGG	Carbon metabolism	5	15.9	2.0e-04	0.012
age_down	KEGG	Lipoic acid metabolism	3	59.8	1.0e-03	0.030
age_down	KEGG	Citrate cycle (TCA cycle)	3	35.9	2.8e-03	0.041

List	Category	Term	Count	fold	p	FDR
age_down	GO-BP	tricarboxylic acid cycle	4	50.2	6.2e-05	0.012
age_down	GO-BP	malate-aspartate shuttle	3	142.3	1.8e-04	0.018
age_down	GO-BP	mitochondrial fission	3	85.4	5.2e-04	0.034
age_up	GO-BP	mRNA splicing, via spliceosome	4	10.0	6.7e-03	0.960
age_up	GO-BP	RNA splicing	4	9.1	8.6e-03	0.960
age_up	GO-BP	regulation of neurogenesis	2	27.4	6.8e-02	1.000
cluster1_up	KEGG	EGFR tyrosine kinase inhibitor resistance	3	10.1	3.3e-02	1.000
cluster1_up	KEGG	Pathways in cancer	6	3.0	4.0e-02	1.000
cluster1_up	KEGG	Ras signaling pathway	4	4.6	5.1e-02	1.000
cluster1_up	GO-BP	lung development	4	25.7	4.8e-04	0.290
cluster1_up	GO-BP	embryonic organ development	3	24.1	6.6e-03	1.000
cluster1_up	GO-BP	in utero embryonic development	3	18.3	1.1e-02	1.000
cluster2_up	KEGG	Polycomb repressive complex	5	10.7	1.1e-03	0.092
cluster2_up	KEGG	Endocytosis	6	4.3	1.2e-02	0.500
cluster2_up	KEGG	Spliceosome	4	5.0	4.3e-02	1.000
cluster2_up	GO-BP	mRNA splicing, via spliceosome	6	5.2	5.7e-03	1.000
cluster2_up	GO-BP	multivesicular body assembly	3	16.5	1.4e-02	1.000
cluster2_up	GO-BP	lung development	3	15.1	1.6e-02	1.000
SI_top200	KEGG	Breast cancer	5	4.0	3.5e-02	1.000
SI_top200	KEGG	Endocrine resistance	4	4.8	5.0e-02	1.000
SI_top200	GO-BP	positive regulation of cell cycle process	3	33.3	3.4e-03	1.000
SI_top200	GO-BP	determination of left/right symmetry	4	12.0	4.4e-03	1.000
SI_top200	GO-BP	pH reduction	2	74.1	2.7e-02	1.000

```
sig <- dv[dv$FDR < 0.05, c("Category", "Term", "Count", "Genes", "FDR")]
sig$FDR <- signif(sig$FDR, 2)
knitr::kable(sig, row.names = FALSE,
             caption = "All DAVID terms with FDR < 0.05 (age_down list) and the genes that drive
```

Table 11: All DAVID terms with FDR < 0.05 (age_down list) and the genes that drive them.

Category	Term	Count	Genes	FDR
KEGG	Carbon metabolism	5	GCSH, GOT1, MDH2, DLAT, PDHB	0.012
KEGG	Lipoic acid metabolism	3	GCSH, DLAT, PDHB	0.030
KEGG	Citrate cycle (TCA cycle)	3	MDH2, DLAT, PDHB	0.041
KEGG	Metabolic pathways	11	GCSH, ALAS1, GOT1, MDH2, CA4, CISD1, L2HGDH, DLAT, PDHB, NDUFV3, DCTPP1	0.041
KEGG	2-Oxocarboxylic acid metabolism	3	GOT1, DLAT, PDHB	0.041
GO-BP	tricarboxylic acid cycle	4	MDH2, L2HGDH, DLAT, PDHB	0.012
GO-BP	malate-aspartate shuttle	3	GOT1, MDH2, SLC25A12	0.018

Category	Term	Count	Genes	FDR
GO-BP	mitochondrial fission	3	MUL1, MTFR1L, MFF	0.034

Interpretation. Only the age-decreasing list gives terms that survive FDR < 0.05, and they form one coherent theme: mitochondrial energy metabolism. KEGG carbon metabolism, lipoic acid metabolism and the citrate (TCA) cycle, and GO-BP tricarboxylic acid cycle, malate-aspartate shuttle and mitochondrial fission are all enriched, driven by a small core of genes (DLAT and PDHB of pyruvate dehydrogenase, MDH2, GOT1, GCSH, L2HGDH, SLC25A12, and the fission genes MFF, MUL1 and MTFR1L). This is the "mitochondrial dysfunction" hallmark of aging and supports the mitochondrial part of H3, but not its extracellular-matrix or inflammation parts, which are not significantly enriched in either age list. The genes increasing with age show only a nominal mRNA-splicing signal (FDR about 0.96). The two cluster marker lists give only nominal terms (cluster 1: growth-factor receptor signaling through PDGFRB, FGFR2 and VEGFA; cluster 2: Polycomb repressive complex, splicing and endocytosis, best FDR = 0.09), which matches the weak cluster structure (silhouette about 0.02). The SI list has no pathway-level signal, so the SRF result above rests on the targeted gene tests, not on DAVID.

9 Discussion

Main result: the muscle transcriptome has little strong, unsupervised structure. The heatmap of the 1,000 most variable genes shows no clear sample blocks, average silhouette widths are about 0.02 for every k from 2 to 6 (values near 0 mean no real cluster separation), and k-means and hierarchical clustering only partly agree (ARI = 0.57). The best k-means split (k = 2) is not enriched for sex, group, diabetes, or age after FDR correction; the strongest nominal signal is insulin sensitivity (SI, p = 0.033) with BMI and fasting insulin close behind. H1 is therefore only weakly supported: whatever separates the clusters is related to insulin resistance / adiposity rather than to the recorded diagnostic groups.

Unexpected finding 1: sex is not the dominant axis (H2 rejected). Only 19 genes differ strongly by sex, almost all on the X/Y chromosomes (XIST, TSIX, RPS4Y1, DDX3Y, KDM5D, UTY, USP9Y, ZFY, EIF1AY), plus a few autosomal adipose-related genes (LEP, ADH1B, CD24), which probably reflect sex differences in intramuscular fat. With ~22,000 genes these 19 genes contribute almost nothing to PCA or clustering, so removing them changed neither result. Men and women do not form distinct clusters.

Unexpected finding 2: data-quality problems. Two of 50 samples disagree with their reported sex (P25 looks male; N1 expresses both XIST and Y genes and also has an implausible WHR), and two samples (P1, D10) are global outliers that drive the leading PCs. In the full PCA, PC1 explains only 7.6% of the variance and its top loadings (ribosomal, hemoglobin, heat-shock genes) look like tissue-composition or RNA-quality effects rather than biology. After removing these samples and scaling genes, PC2 becomes associated with SI (p = 0.018) and fasting insulin, which matches the clustering result.

Age-associated genes (H3, H4). No gene passes FDR < 0.05 for age (the best q is about 0.05), which is expected with n = 50 and a 21 to 67-year age range, so the top 100 genes by p-value are used as an exploratory set. Their direction is informative: many of the strongest age-*decreasing* genes encode mitochondrial and energy-metabolism proteins (MFF, mitochondrial fission; DLAT, pyruvate

dehydrogenase; MDH2, TCA cycle; ATP5F1, ATP synthase; MRPL15, mitochondrial ribosome), which fits the "mitochondrial dysfunction" hallmark of aging. However, the correlations weaken in the 40 non-diabetic subjects and only 1 gene (UNC13C) remains significant after adjusting for sex and diabetes. Because the T2D group is about 12 years older, part of the apparent age signature is a diabetes signature, so H4 is supported. DAVID confirms that the mitochondrial theme is statistically enriched in the age-down list (KEGG carbon metabolism FDR = 0.012, GO-BP TCA cycle FDR = 0.012), but because the input list is exploratory and confounded with diabetes, this shows *where* the age/diabetes signal concentrates rather than proving an age-only effect.

Reproduction of Jin et al. SRF and its co-activator MKL1 are negatively correlated with insulin sensitivity (SRF r = -0.37, MKL1 r = -0.44), and SRF and the SRF targets VCL and CNN2 are higher in T2D than in FH- subjects (nominal $p < 0.05$). This agrees with the paper's conclusion that higher SRF activity marks insulin resistance, although none of these survive genome-wide FDR on their own.

Approaches that did not work, and why. (1) Clustering on all variable genes produced clusters with silhouette near 0, because between-subject biological differences are small compared to array noise (median inter-sample r = 0.874). (2) Removing sex genes did not reveal hidden structure, because sex was never the main driver. (3) Strict FDR filtering gave no age genes, so a rank-based list had to be used, and pathway results from it should be called exploratory. A better design would model age and glycemic status jointly, use a larger cohort, or use gene-set methods (GSEA) that pool weak signals across many genes.

Limitations. Single time point, small groups (10 T2D), confounding between age and diabetes, missing OGTT data for all T2D subjects, and two suspicious sample records.

10 AI usage

Claude (Anthropic) was used to draft the analysis plan, write the R code in this R Markdown file, and draft the text. The analysis choices (gene filtering, sex-gene removal, confounder adjustment, outlier handling) and all interpretations were reviewed and checked against the output by the author. Code was run in a GitHub Codespace / GitHub Actions runner (rocker/verse container). The DAVID queries were submitted through the DAVID website by Claude using browser automation; the returned tables are stored unchanged in `data/david/`.

11 References

1. Jin W, Goldfine AB, Boes T, et al. Increased SRF transcriptional activity in human and mouse skeletal muscle is a signature of insulin resistance. *J Clin Invest.* 2011;121(3):918-929. GEO: GSE25462.
2. López-Otín C, Blasco MA, Partridge L, Serrano M, Kroemer G. The hallmarks of aging. *Cell.* 2013;153(6):1194-1217.
3. Lu T, Aron L, Zullo J, et al. REST and stress resistance in ageing and Alzheimer's disease. *Nature.* 2014;507:448-454 (GDS5204, extra-credit brain dataset, not analyzed here).


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## [1] "R version 4.4.1 (2024-06-14)"
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